

Coal Assignment 03

Name: Yash Raj

Roll no: 24K-01737

Ques:-

INCLUDE Irvine32.inc

.data

Initial Value DWORD 0D4A4h

Divisor DWORD 0Ah

finalResult DWORD 0

.code

main PROC

INVOKE RepeatedDivide, InitialValue, Divisor

mov eax, finalResult

call WriteHex

call crlf

exit

main ENDP

RepeatedDivide PROC uses ebx ebx
Number: DWORD, factor: DWORD

mov eax, Number

cmp eax, 5h

jbe finished

Name: Yash-Raj

Roll-no: 24K-07-51

```
mov ebx, factor  
xor ecx, ecx  
div ebx
```

INVOKE RepeatedDivide, ecx, factor.

finished:

```
mov finalResult, ecx  
ret
```

RepeatedDivide ENDP

END main

Dr.

Name: Yash Raj

Roll-no: 21K-0737

Q12.

INCLUDE Trini32.inc

.data

Msgt byte "Enter a string:", 0

input byte 200 DUP(0)

vowelA count Dword 0

vowelB count Dword 0

vowelI count Dword 0

vowelO count Dword 0

vowelU count Dword 0

vowelHeader byte "Vowel Count:", 0

labelA byte "a or A's", 0

labelE byte "e or E's", 0

labelI byte "i or I's", 0

labelO byte "o or O's", 0

labelU byte "u or U's", 0

.code

main PROC

call CountVowels

exit

main ENDP

To be continued....

Name: Yash-Ray
Continued here:

Roll-no: 2410-0737

CountVowels PROC

mov ecx, offset my1
call WriteString

mov ecx, offset inputbuffer
mov ecx, 200
call ReadString

mov esi, offset inputbuffer

next character:

mov al, [esi]

cmp al, 0

je finishedCounting

cmp al, 'A'

je addA

cmp al, 'a'

je addA

cmp al, 'E'

je addE

cmp al, 'e'

je addE

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Roll no: 24K-0737

cmp al, 'i'

je addI

cmp al, 'i'

je addI

cmp al, 'o'

je addO

cmp al, 'o'

je addO

cmp al, 'u'

je addU

cmp al, 'u'

je addU

jmp skipchar

inc vowelA-count

addA: jmp skipchar

addB: inc vowelB-count

jmp skipchar

addI: inc vowelI-count

jmp skipchar

addO: inc vowelO-count

jmp skipchar

addU: inc vowelU-count

jmp skipchar

Name: Yash-Raj

Skipchar:

inc esi

jmp nextcharacter

finished counting:

call crlf

mov ecx, offset vowelCharacter

call WriteString

call crlf

mov ecx, offset labelA

call WriteString

mov ecx, vowelA-Count

call WriteDec

call crlf

mov ecx, offset labelE

call WriteString

mov ecx, vowelE-Count

call WriteDec

call crlf

mov ecx, offset labelI

call WriteString

mov ecx, vowelI-Count

call WriteDec

call crlf

Name: Yash Raj

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mov ecx, offset label0

call WriteString

mov ecx, vowel-count

call WriteDec

call ctf

ret

CountVowels ENDP

END main

Am

Name: Yash-Raj

Roll-no: 2410

Q#3

INCLUDE Irvine32.inc

.data

str byte '###FAST', 0

.code

main PROC

~~Strip~~ INVOKE stripleading, ADDR str, '#'

mov ecx, offset str

call WriteString

call crlf

exit

main ENDP

stripleading PROC uses esi, ecx

extrn PTR byte, charToDrop: byte

mov esi, extrn PTR

mov al, charToDrop

checkstart:

mov bl, [esi]

cmp bl, 0

je startshift

cmp bl, al

jne startshift

inc esi

jmp checkstart

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Roll no: 24K-4337

startlib:

mov edi, bufP1

CopyBuf:

mov bl, [esi]

mov [edi], bl

cmp bl, 0

je donestrip

inc esi

inc edi

jmp CopyBuf

donestrip:

ret

StripTrailing ENDP

END main

Roll-no. 24K-0737

Name: Yash Raj

Q#42)

INCLUDE Irvine32.inc

• data

arr1 Dword 5, 5, 5, 5, 5, 9, 1

arr2 Dword 1, 5, 5, 5, 5, 4, 5

arr3 Dword 5, 5, 5, 0, 5, 5, 5, 5, 5

• code

main PROC

INVOKE findfive, offset arr1, LENGTHOF arr1

call WriteDec

call crlf

INVOKE findfive, offset arr2, LENGTHOF arr2

call WriteDec

call crlf

INVOKE findfive, offset arr3, LENGTHOF arr3

call WriteDec

call crlf

exit

main ENDP

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Roll-no: 24K0737

findfive PROC cuxi ebx, ecx edx esi,
pArr: PTR DWORD, size: DWORD

mov esi, pArr

mov ecx, size

mov eax, 0

Scanloop:

cmp ecx, 5

jl noMatch

mov edx, 5

mov ebx, 5

mov edi, esi

checkRun:

mov eax, 5edi

cmp eax, ebx

jne nextStart

dec edx

cmp edx, 0

je found

add edi, 4

jmp checkRun

Roll-no 3241-17

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nextstart:

add esi, 4

dec ecx

jmp scanloop

found:

mov ecx, 1

jmp finish

nomatch:

mov ecx, 0

finish:

ret

finclfive ENDP

END main